Everything, in one place

Marcus — The Complete **File**.

His course plan, career roadmap, the grants, the entrepreneur's playbook, and the reminder of what he's already won — one document, one consistent design.

Prepared for **Marcus's dad** · University of Guelph · Biological & Pharmaceutical Chemistry · Class of 2030 · June 2026

How to use this file

This single document holds **everything we built together** — seven chapters, front to back. No more hunting through the chat. Read it in any order; each chapter opens with its own title page.

> IF YOU ONLY HAVE FIVE MINUTES
>
> **Hand Marcus Chapter 2 — the one-pager (p. 11).** It's the "you've already won, three for three" reminder, written for him to read in two minutes.
>
> **Then skim Chapter 3, the Course Plan (p. 13)** — plain English on what to take and when. That's the practical core.
>
> The remaining chapters are depth — the detailed degree map, careers and companies, the founder's path, the grants, and the big-picture life roadmap — there whenever you want them.

What's inside

Chapter One

The Proof.

For Marcus — what you've already won, and why the fire isn't gone. It's just resting.

A letter to you

Read this part slowly.

Marcus — this isn't a lecture, and it isn't a plan with a hundred steps. It's just the truth, laid out plainly, because sometimes when you're in the fog you can't see the thing that's obvious to everyone standing around you.

You've been told you're losing your drive. I don't think that's what's happening. I think you've been running hotter, for longer, than almost anyone your age ever has — and what you're feeling right now isn't the fire going out. It's the cost of how hard it's been burning. There's a difference, and it changes everything.

The ledger

Look at what's already behind you — the hardest things a young person can be handed.

WIN №1 — THE ONE MOST PEOPLE NEVER FACE

You had cancer. And you beat it. Most people will live their whole lives and never be tested the way you already have been — and you came through it. That is the hardest thing there is, and it's behind you.

WIN №2 — AND THEN YOU DIDN'T COAST

You took on the hardest year of your life — and finished it. You'd have been forgiven for taking it easy after everything. You didn't. You walked straight into the toughest academic year you'd ever had and saw it through.

WIN №3 — YOU AIMED ABOVE YOUR OWN EXPECTATIONS

You got into a program you thought might be out of reach. You set the bar higher than you believed you could clear — and you cleared it. Three for three isn't luck. It's a **pattern**.

Recovery, not regression

What this season actually is.

The flatness, the drained tank, the *"where did my drive go"* — that is not you slipping backward. That is what happens to a person **after** they win a long, hard fight.

Your body and mind ran in survival mode for a long time. They're allowed to exhale now. Soldiers come home and crash. Athletes who win the championship feel empty the week after. It isn't the absence of fire — it's the bill for having burned so hot, for so long.

> THE WHOLE THING IN ONE LINE
>
> You are not **regressing**. You are **recovering**. Those are completely different things — and only one of them is a problem. Yours isn't.

How the fire actually comes back

You can't force it — you've been through too much for shame to work as fuel.

Evidence

Remembering you've already won. That's what the ledger is — proof in writing.

Small wins

One thing, slightly hard, that you finish. Momentum rebuilds one rep at a time.

Rest

The best periodize — push, recover, push harder. Rest isn't quitting. It's training.

The pattern you've already proven

Can you win over and over? You already are.

Beating cancer was a win. Finishing that year was a win. Getting in was a win. And every one of them was **harder** than the next thing standing in front of you.

First-year physics? You've survived chemo. A brutal semester? You've already done a tougher one. You are not *hoping* you can do hard things. You have a **track record**.

> SO WHEN THE NEXT HARD THING COMES
>
> It isn't a test of whether you've got it. It's just the next rep — and **you have already lifted heavier.**

If you want a place to start

Completely optional. Pick one. Finish it. That's the whole assignment.

- **Move your body once a day** — a walk counts. Remind your system it's safe to come back online.
- **Finish one small thing you've been avoiding** — a single email, a drawer, a form. Completion is a drug. Take the dose.
- **Reconnect with one person** — a friend, not a network move. Being seen is fuel.
- **Read four pages of anything** — just to feel your mind turning over again, at its own pace.
- **Sleep like it's your job** — because for now, it is. Recovery happens there.

If the fog is heavy

One honest thing, from someone in your corner.

If the flatness runs deeper than a slow summer — if it's been hanging on, if things you used to care about feel grey — that is **incredibly common** for people who've come through what you have. The fight doesn't end when treatment does; sometimes the hardest part is the quiet after.

Reaching for support isn't a crack in the armor. The elite athletes you'd respect all have a team behind them. Your care team almost certainly has survivorship and counselling people whose entire job is exactly this. Using them is the same smart strategy you've used to win everything else: **get the right people in your corner, then go win.**

A quiet look ahead

It's there when you're ready. Not a day before.

> There's a whole map of what's ahead if you ever want it — the program, the career, the doors that open one after another. Your dad has it. But you don't need it today. Today the only job is this: **rest like you've earned it** — you have — and when you're ready, pick one small win. The big things will still be there, and you'll walk into them the same way you've walked into everything else, and come out the other side.

You've already proven who you are.
Nobody's asking you to become someone new —
just to remember someone you already are.

Chapter Two

The One-Page Version.

The whole message at a glance — the page to actually hand him.

You've already won the hardest games of your life.

The proof — three for three

01	You had **cancer**. And you beat it. The hardest thing there is — behind you.
02	Then you took on the **hardest year of your life** — and finished it.
03	And you got into a **program you thought was out of reach**. You aimed high and hit it.

> What you're feeling right now isn't your fire going out. It's the cost of how hard it's been burning. You're not **regressing** — you're **recovering**. Those are completely different things.
>
> The best in the world don't grind year-round. They push, recover, then push harder. Right now, rest isn't quitting. It's **training**.

First-year physics? **You've survived chemo.** A brutal semester? You've already done a tougher one. The next hard thing isn't a question mark — it's just the next rep, and **you've already lifted heavier.**

So the only job today: rest like you've earned it, and when you're ready, pick **one** small win. Finish it. Then the next. That's how the fire comes back — one rep at a time.

You've already proven who you are.
Just remember someone you already are.

Chapter Three

The Course Plan.

Plain-English: what to take each semester, and the reasoning behind each choice.

The plan, in plain English

Built around what he's optimizing for: performance, money, and enjoyment

$50–80K	**$80–95K**	**$110K+**
EARNED ACROSS 4 CO-OP TERMS	STARTING SALARY	BY ~30, PLUS A PENSION

THE WHOLE PLAN IN FOUR SENTENCES

1. Survive first-year physics — two courses, and his grades there decide whether he stays in co-op.

2. Years 2–3 mostly run themselves — the program picks his chemistry courses for him.

3. From Year 2, aim every elective at his target field (nuclear/radiopharma): inorganic, environmental, instrumental analysis.

4. Quietly stack a few business courses so he graduates looking like a future leader, not just a chemist.

The one thing to watch: physics

He didn't take physics in high school — so he's on the gentler track.

- **PHYS*1300** (Fall, Yr 1) — the physics course built for students without Grade-12 physics. Algebra-based, the friendlier path.
- **PHYS*1080** (Winter, Yr 1) — the follow-up. After this, **no more physics, ever.**

WHY THIS MATTERS MORE THAN ANYTHING ELSE IN YEAR 1

The co-op program requires a **70% average after second semester** — exactly when both physics courses finish. A bad grade there can knock him out of co-op and collapse the whole earnings plan. **Tutoring from week one**, and bank a GPA cushion in the lighter first year.

The map, year by year

WHEN	COURSES	WHY THIS SET
Yr 1 · Fall	CHEM*1040, BIOL*1090, MATH*1200, **PHYS*1300**, + ECON*1050	First-year science + catch-up physics. Econ is the first management seed.
Yr 1 · Winter	CHEM*1050, MATH*1210, **PHYS*1080**, BIOL*1070, COOP*1100, + PSYC*1000	Finish physics. COOP*1100 preps for work terms. Psych = easy high mark + people skills.
Yr 2 · Fall	CHEM*2060, CHEM*2400, CHEM*2700, STAT*2040	The hardest semester — all locked. CHEM*2400 (Analytical) lands the first lab job.
Yr 2 · Winter	WORK TERM 1	Government or university lab — résumé credibility for bigger roles next.
Yr 2 · Summer	CHEM*2480, BIOC*2580, MICR*2420, + BUS*2220	Commit to the pathway. BUS*2220 keeps building the leadership story.
Yr 3 · Fall	BIOC*3570, CHEM*3430, **CHEM*3640, ENVS*3030**	The key elective choice — inorganic + environ-mental = field-relevant signal.
Yr 3 · Winter	CHEM*3750, **CHEM*3650, CHEM*4020**, + MGMT*3020	More field-relevant chemistry + a 4th management course.
Yr 3 Su–Yr 4 F	WORK TERMS 2 & 3 (back-to-back, 8 months)	The leverage block — one employer, one real owned project.
Yr 4 · Winter	4000-level CHEM + **MGMT*4000** + electives	Top-level chemistry + the capstone management course.
Yr 4 · Summer	WORK TERM 4	Convert it into a full-time offer before graduating.
Yr 5 · Fall	Final 4000-level + electives	Light final term — ideally job-search-free with an offer in hand.

Confirm the exact grid with a BPCH faculty advisor each year — Guelph reshuffles course placement annually.

Chapter Four

The Course Map.

The detailed degree map — every choice point, scored on performance, money, and enjoyment.

Where he actually has a choice

Most of BPCH is prescribed — the strategy lives in a few decision points

At the course level, ~85% of the degree is fixed. The program picks it for him. The real optimization is in five places — and they don't begin until Semester 4.

DECISION	THE CALL, AND WHY
Sem 1 physics LOCKED	PHYS*1300 — forced by no Grade-12 physics, and it's the gentler track. Two courses, Year 1 only, then done.
Sem 2 biology	**BIOL*1070** over 1080 — historically easier-graded, protects the 70% gate. Biology isn't load-bearing for his path.
3000/4000 restricted electives HIGH IMPACT	The main lever. **CHEM*3640/3650 (inorganic) + ENVS*3030 (environmental) + CHEM*4020 (instrumental)** — exactly what nuclear/radiopharma employers want to see.
Liberal-Ed electives	Spend them as management seeds — **Econ, Psych, Org Behaviour, Strategic Management**. Reliable high marks + a "future leader" transcript no other chemist has.
CHEM*4900 research thesis	Take it *only* if grad school or a startup idea is live; otherwise lighter electives protect GPA.

THE HONEST ONE-LINER

Follow the prescribed sequence on autopilot until Semester 4 — then steer every elective toward the field he commits to, and stack the management courses nobody else thinks to take.

The decision register

Each choice, scored — Performance · Money · Enjoyment

CHOICE	PICK	WHY
Pathway	**Nuclear / radiopharma** (see Ch. 5)	Highest baseline + clearest ladder + the field that unites his interests
Sem 2 biology	BIOL*1070	GPA protection at the gate
Sem 5/6 electives	Inorganic + environmental + instrumental	Turns his transcript into a hiring magnet
Liberal-Ed	Management sequence	High marks now, leadership signal later
CHEM*4900	Skip unless founder/grad-school	Protects GPA for industry; take it only with intent

Full risk register and operational checklist live in the original detailed map; the essentials are captured above.

Chapter Five

The Inside Track.

Who to meet, what companies are out there, how to get involved in startups — and the one frontier that unites everything.

One frontier unites everything

Where his nuclear pull, his pharmaceutical-chemistry degree, his drive — and his story — meet

RADIOPHARMACEUTICALS — THE UNIFYING FIELD

Radiopharmaceuticals attach **radioactive isotopes to drug molecules** to deliver radiation straight to cancer cells. It is **nuclear chemistry and pharmaceutical chemistry fused into one** — and Canada is a world leader. For Marcus it collapses every fork into a single, fundable, world-class path.

THE ROLE MODEL — STUDY THIS STORY

Dr. John Valliant, a McMaster chemistry professor, spun his research out of the university's radiopharma centre (the **CPDC**) into **Fusion Pharmaceuticals** — radioconjugates to treat cancer.

In June 2024, **AstraZeneca acquired Fusion for ~$2.4 billion USD.** A chemist. A cancer therapy. A two-billion-dollar outcome, built in Ontario from a university lab.

And it should land for Marcus in particular: the most fundable, meaningful thing a chemist can build right now is the kind of targeted therapy that saves people like him.

The people to meet

Real, current Guelph chemistry faculty — start these relationships in Year 1

PROFESSOR	WHY HE SHOULD KNOW THEM
Dr. Mario Monteiro	**Target first.** Builds carbohydrate vaccines; his *Campylobacter* vaccine reached human trials and is WHO-listed. A Guelph chemist turning lab work into real medicine.
Dr. Richard Manderville	Bioorganic / medicinal — DNA damage, aptamers, probes. Core drug-discovery science.
Dr. France-Isabelle Auzanneau	Carbohydrate & medicinal — tumour-associated antigens (cancer-relevant). Award-winning.
Dr. Derek O'Flaherty	Nucleic-acid / RNA chemistry — the tech behind RNA & mRNA drugs.
Drs. Schwan & Huang	Organic synthesis & methodology — the craft of making any drug molecule.

Tactic: take their courses, go to office hours in Year 1, and be in a lab by Year 2. A professor's reference and lab spot is the highest-leverage relationship an undergrad can build.

The companies — the whole map

TIER	COMPANIES (NAMED)
Big Pharma (Ontario)	AstraZeneca, GSK, Roche, Amgen, Teva, Sanofi, Bayer, Apotex, Pharmascience
Drug-discovery CROs	Dalriada (Ontario's largest), Sygnature Discovery, Charles River Labs
CDMOs / manufacturing	Catalent, Dalton Pharma Services, ITR Labs, Applied Pharmaceutical Innovation
Radiopharma HIS FRONTIER	Fusion (now AstraZeneca), ARTMS, AtomVie, Canprobe, CPDC; Laurentis (OPG isotopes)
Biotech startups	Deep Genomics, BlueRock Therapeutics, ProteinQure, BenchSci
The cluster / front doors	MaRS Discovery District, JLABS @ Toronto, CCRM, OmniaBio

How a scientist becomes a funded founder

The machinery nobody teaches science students — in the order you use it

- **At Guelph:** John F. Wood Centre → **IgniteLab** → **LaunchLab** (up to $5K, mentorship). Low stakes — the point is reps.
- **After a degree/thesis: Lab2Market** (Discover → Validate → Launch, $15K Mitacs grants) → **Creative Destruction Lab** (elite accelerator, investor exposure). Also Velocity, Next Canada.
- **The radiopharma shortcut:** a master's/PhD at McMaster's **CPDC** — the machine that built Fusion — is arguably the highest-probability path in Canada for a chemist to become a venture-backed founder. Valliant did exactly this.

FUNDING (NON-DILUTIVE FIRST)	WHAT IT GIVES
Mitacs Accelerate	Subsidized grad/postdoc researchers (~$15K units) — easiest entry
NRC IRAP	Up to **$1M** for R&D salaries; a dedicated advisor; rolling intake
Innovative Solutions Canada	Phase 1 up to **$150K**; Phase 2 up to **$1M**
NSERC Alliance + SR&ED	2:1 research match + R&D tax refunds — stack them all

~$5B/yr flows through these programs, and they accept pre-revenue companies. The edge is simply knowing they exist.

Sources: U of Guelph Dept. of Chemistry faculty; AstraZeneca–Fusion acquisition (~$2.4B, June 2024); CPDC/McMaster; MaRS, JLABS, CCRM, Dalriada; U of G John F. Wood Centre; Lab2Market (Mitacs); NRC IRAP; Innovative Solutions Canada. Names are research/relationship targets — confirm current faculty & openings directly.

Chapter Six

Money & Momentum.

The grants he qualifies for with deadlines, the hiring edge most students never use, and where the salary actually goes.

The grants — with deadlines

Honestly filtered: what he qualifies for, and what to skip

GRANT	AMOUNT	ELIGIBLE?	WHEN TO APPLY
NSERC USRA	~$6,000	**YES**	**Jan–Feb 2027** for that summer, via the Guelph dept
OEN–Bruce Power Nuclear Studies	$3,500	**YES**	Opens ~Feb–Mar; first shot **Spring 2027**
Canadian Nuclear Society research	~$7,000	**YES**	After 2nd year; needs a supervisor. Join CNS (free) now
EHRC "Empowering Futures"	$5–7K *(to employer)*	**YES**	Any work term — he names it to be cheaper to hire
Bruce Power $500	$500	**MAYBE**	May 15–Jun 15 — only if from Bruce/Grey/Huron county
OPG in-course awards	varies	**MAYBE**	Check Guelph awards office
WiN Canada	$3,000	**NO**	Women only — not eligible

DO THESE TWO THINGS NOW

1. Have him **join the Canadian Nuclear Society** (free). 2. Calendar **NSERC USRA (Jan–Feb 2027)** and the **OEN–Bruce Power $3,500 (Spring 2027)** as the two real targets.

The hiring edge most students never use

Make employers want him — because he's partly free

The federal **EHRC "Empowering Futures"** program pays an *employer* up to **$5,000** (or **$7,000** for a first-year student) to take a student in an energy-sector work term. When Marcus applies — especially to a smaller lab — he says: *"hiring me is partly subsidized through EHRC Empowering Futures."* It makes him the easy yes. Almost no 19-year-old knows this.

Where the salary actually goes

$80K out of school — and the real ladder from there

ROUTE	PATH & TOP END
Standard — technical	Lab/station chemist → senior → supervisor. **$110–140K** by ~30 + a pension worth ~$1.3M.
Superior — leadership	P.Eng + MBA → director/VP or consulting. **$180K–$1M/yr**; multi-million net worth.
Elite — founder / C-suite	Radiopharma founder or executive. Verified: OPG CNO **$1.09M**, CEO **$1.6M**; a radiopharma exit hit **$2.4B**.

THE HONEST VERSION

The $80–95K start is **near-certain** if he executes. The millions are **earned, not guaranteed** — they come from the credentials, the niche, and the network. But the floor is high and the ceiling is real. Few degrees can say both.

Chapter Seven

The Atlas.

The three-lives roadmap — Standard, Superior, Elite — and the one rule that ties them together.

The one rule

How a careful man and an ambitious man can be the same man

> These are **not three roads you choose between at eighteen.** They are one path, where each tier is an **option the previous tier unlocks.** You build the Standard foundation — it creates the option on Superior; Superior creates the option on Elite. **You never burn the floor to chase the ceiling.**

The three tiers, side by side

	STANDARD	SUPERIOR	ELITE
The life	Partner, two kids, secure, upper-middle-class	The expert others hire; real influence	Shapes the industry; sits at the table
Vehicle	Chemist → senior → manager	P.Eng + MBA → consulting / VP / advisory	Founder · C-suite · policy–capital
Money	$110–180K + pension	$300K–$1M+/yr; $3–8M net worth	$1M+/yr; equity & influence
Odds if he executes	**~85%+**	**~25–40%**	**single digits**

What the council says

A few of the fifteen voices, on the life he's building

> ### The Wealth Strategist
> The Standard life's secret is that the **floor compounds.** Get him to his first $100K invested by 26 — that's the hardest dollar. Everything after accelerates.

> ### The Nuclear Veteran
> The risk in this tier isn't money — it's comfort. Make the bigger moves in his **late twenties, before the pension handcuffs close.**

The Life Architect

A partner, two kids, work that matters — that's what most of humanity strives toward. If he lands here and is happy, that is not failure. Standard is the net that lets him swing **without fear.**

The Ex-MBB Partner

A P.Eng who's done real chemistry and then gets an MBA is a unicorn. Be the person who speaks **both** the chemistry and the money — that person is never cheap.

The Executive Coach

Pursue Elite as a possibility, not a verdict on your worth. Build the platform, maximize the surface area for luck, and let the floor keep you sane if the biggest door never opens.

The spine — by age

AGE	THE GATE, AND THE OPTION IT BUYS
18–22	Clear the GPA gate. Co-ops to the frontier. Grants. SECURES STANDARD · BUYS SUPERIOR OPTION
22–27	Master a niche. P.Eng file. First $100K. Build the reputation. SHARPENS SUPERIOR
27–32	MBA / consulting / independent — Superior is won here, before the handcuffs. BUYS ELITE OPTION
32+	Platform → the table: partner, VP, founder, board, policy. ELITE, IF DOORS OPEN

The recurring loop

The self-test he runs for life

- **Each semester:** Did I hold the gate? Is the next co-op pointed at the frontier?
- **Each year:** Is the floor secure? Did I take one action that buys an option on the next tier?
- **Every ~3 years:** Re-rank the tiers against who I've actually become, and re-decide.

THE LAST WORD

Marcus — you're not choosing a job. You're positioning yourself to enter a short-staffed, high-paying industry in its biggest growth decade, with a secure life nearly guaranteed and **two ladders climbing toward the extraordinary.** Build the floor. Buy the options. Run the loop. And let the doors open in their own time.

Verification: Guelph co-op & Sunshine List data (OPG President $1.91M, CEO $1.60M, CNO $1.09M); MBB 2026 pay guides; Professional Engineers Ontario; SMR funding ($3B Darlington, X-energy $1.8B); AstraZeneca–Fusion $2.4B. Career ranges beyond entry are estimates and vary. A strategic framework, not a guarantee. Prepared June 2026.